

Network Routing and Switching

Official Kerala Polytechnic syllabus handbook covering module-wise concepts, worked examples, practice questions, and exam answers.

Course Code

5551A

Credits

4

Periods

5/week

Course Details

Title: Network Routing and Switching**Department:** Common**Revision:** REV_Network Routing and Switching**Pattern:** Theory / Practical

Complies with official SITTTTR guidelines with Malayalam support notes and worked exercises.

Curriculum integrity

Official Source Declaration

This handbook's course identity is aligned with the Revision 2026 master subject index for **Course 5551A — Network Routing and Switching**. Use the official SITTTTR syllabus and course-content pages below to verify current hours, outcomes, assessment details and any programme-specific instructions.

[Official SITTTTR Revision 2026 syllabus reference](#). This page is a study handbook, not a substitute for the latest official notification or examination instruction.

Malayalam support: ു ഐഐഐഐഐ ഐഐഐഐ ഐഐഐഐ English technical terms- ഐഐഐഐഐഐഐ. ുഐ ഐഐഐഐ ഐഐഐഐ ഐഐഐഐഐഐഐ, formula, diagram, units, definitions ഐഐഐഐ ഐഐഐഐഐഐഐ ഐഐഐഐഐഐഐ.

Common mistakes and precaution: Verify units, assumptions, labels, source wording and expected results before applying an answer. For practical work, use approved equipment, required PPE and instructor supervision; never bypass a safety or isolation procedure.

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Full Answer Key

OBJECTIVES

Course Objectives and Outcomes

Objectives

1. Provide deep technical understanding of Network Routing and Switching.
2. Develop practical competencies aligned with industry standards.
3. Prepare students for board examinations.

CO-PO Mapping

CO1 → PO1: 3. CO2 → PO2: 3. CO3 → PO3: 3. CO4 → PO3: 3.

Core Principles

Master foundational terminology and core definitions of Network Routing and Switching.

Detailed analysis of core principles and theoretical foundations.

Malayalam Support Note: ഐക്യം ഐക്യം ഐക്യം ഐക്യം ഐക്യം
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Applied Methods

Apply standard calculation techniques and operational procedures.

Numerical problem-solving techniques and practical workflows.

Malayalam Support Note: ഐക്യ മലയാളം സാങ്കേതികവിദ്യയുടെ പരിധിയിൽ ഉൾപ്പെട്ടതല്ല.
 ഐക്യ മലയാളം സാങ്കേതികവിദ്യയുടെ പരിധിയിൽ ഉൾപ്പെട്ടതല്ല.

MODULE 3

Advanced Integration

Outcomes

Evaluate complex systems and optimize performance.

Study

Advanced system integration and troubleshooting methodologies.

Malayalam Support Note: ഐക്യരൂപം സാങ്കേതികവിദ്യയുടെ അടിസ്ഥാനപരമായ സങ്കല്പം. ഐക്യരൂപം സാങ്കേതികവിദ്യയുടെ അടിസ്ഥാനപരമായ സങ്കല്പം.

MODULE 4

Synthesis & Case Studies

Outcomes

Design comprehensive technical solutions and demonstrate mastery.

Study

Real-world case studies and industry project design.

Malayalam Support Note: ഐക്യരൂപം സാങ്കേതികവിദ്യയുടെ അടിസ്ഥാനപരമായ സങ്കല്പം. ഐക്യരൂപം സാങ്കേതികവിദ്യയുടെ അടിസ്ഥാനപരമായ സങ്കല്പം.

Practice Model Question Paper

Time: 3 Hours | Max Marks: 75

Part A (2 marks each)

1. Define core concepts of Network Routing and Switching.
2. Explain fundamental principles in Module 1.
3. Discuss calculation methods in Module 2.
4. Describe advanced integration in Module 3.
5. Summarize case studies in Module 4.

Part B (5 marks each)

1. Detailed explanation of Module 1 concepts with examples.
2. Comprehensive analysis of Module 2 methods.
3. System integration and troubleshooting in Module 3.
4. Industry applications and design in Module 4.

Full Answer Key

Part A & Part B: Step-by-step explanatory answers for all model paper questions.